

Forbidden Cities

Time limit: 1.00 s

Memory limit: 512 MB

There are n cities and m roads between them. Kaaleppi is currently in city a and wants to travel to city b . However, there is a problem: Kaaleppi has recently robbed a bank in city c and can't enter the city, because the local police would catch him. Your task is to find out if there is a route from city a to city b that does not visit city c .

As an additional challenge, you have to process q queries where a , b and c vary.

Input

The first input line has three integers n , m and q : the number of cities, roads and queries. The cities are numbered $1, 2, \dots, n$.

Then, there are m lines describing the roads. Each line has two integers a and b : there is a road between cities a and b . Each road is bidirectional.

Finally, there are q lines describing the queries. Each line has three integers a , b and c : is there a route from city a to city b that does not visit city c ?

You can assume that there is a route between any two cities.

Output

For each query, print "YES", if there is such a route, and "NO" otherwise.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq q \leq 10^5$
- $1 \leq a, b, c \leq n$

Example

Input	Output
5 6 3 1 2 1 3 2 3 2 4 3 4 4 5 1 4 2 3 5 4 3 5 2	YES NO YES